

MAX SUN

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Education

University of Waterloo

Bachelor of Computer Engineering

2024 - Present

GPA: 3.77/4.0

Experience

Shopify

May 2026 - August 2026

Software Engineer Intern | Ruby, TypeScript, React Native, Android Studio, BuildKite, GraphQL

Toronto, Ontario

- Diagnosed a critical tax breakdown rounding discrepancy by reproducing complex **E2E edge cases**, identifying upstream defects in the **core backend**, and presenting architectural findings to the broader engineering organization.
- Managed the safe deprecation of regional **feature flags** across the **backend** and **mobile POS apps** (iOS/Android) while shipping localization and **UI fixes** for **API endpoints** and email receipt templates.
- Root-caused a gift receipt defect exposing returned items to a **Liquid engine** migration that reused the **giftReceipt** template post-exchange, then aligned with senior engineers on correct partial-return behavior.

BrainRidge Consulting

January 2026 – April 2026

Software Engineering Intern | TypeScript, Node.js, PostgreSQL, AWS, GitHub Actions

Toronto, Ontario

- Designed and deployed a highly-scalable automated HTTP data ingestion pipeline (**Node.js**) for a **RAG-based AI model**, accelerating knowledge base expansion by 70% and **eliminating 83% of manual administrative overhead**.
- Spearheaded the development of a multi-query **RAG-Fusion retrieval engine** utilizing **AWS Bedrock** achieving an unprecedented **98% document recall** for ambiguous user queries.
- Architected a **hybrid search infrastructure** combining **BM25** and **vector search** with **Reciprocal Rank Fusion** within **PostgreSQL**, optimizing exact-match retrieval accuracy and plummeting zero-result rates by 62%.
- Pioneered a real-time, low-latency **conversational AI voice system** via **WebSockets** and **AWS Nova**, delivering sub-500ms median response times for natural banking interactions complete with voice barge-in capabilities.

Exo-Insights

May 2025 – August 2025

Software Developer Intern | Unity, C#, WebGL, Python, Android

Waterloo, Ontario

- Delivered a **digital twin VR training platform** to the **Canadian Nuclear Safety Commission (CNSC)**, building custom file-upload and asset-management systems in **Unity (C#)** for **VR and WebGL**, shipped to production.
- Engineered a **Python-Unity bridge** with JSON-based asset discovery to bypass **WebGL security restrictions**, enabling reliable handling of critical nuclear training files.
- Built a custom **WebGL profiler** that broke a **2-month production bottleneck**, exposing hidden performance issues and becoming standard tooling **across all WebGL projects**.
- Refactored **Android file-loading architecture** delivering **40% faster load times** and reuse across multiple projects.

Projects

Tower of Wizard | Unity, C#

[View Website](#)

- Architected and commercially launched 'Tower of Wizard,' a high-performance 2D action roguelike game (Unity, C#), driving a successful go-to-market strategy that secured **1,000+ wishlists** and **150,000+ impressions**.
- Re-engineered core gameplay physics and logic from standard OOP to **Unity DOTS/ECS** (Entity Component System), unlocking a **500% increase in maximum on-screen entities (10k+)** while slashing frame times by 35%.
- Integrated the **Steamworks SDK** to engineer cross-platform cloud saves, secure achievement tracking, and workshop distribution systems, tangibly **lifting user retention by 12%** within the first month.

Hack the North 2025: EyeCandy | TypeScript, Lens Studio, Gemini

[View GitHub](#)

- Spearheaded the development of an award-winning AR shopping platform (TypeScript, Lens Studio) integrating live **Shopify API** data, securing the **'Runner-Up Best Spectacles Project'** out of **1,000+ hackers**.
- Engineered the spatial interaction layer utilizing full-body tracking, 3D model rendering, and **Google Gemini LLM** voice search, delivering seamless, hands-free product try-ons with sub-2-second end-to-end latency.

Technical Skills

Languages: Python, Java, Kotlin, JavaScript, TypeScript, C++, C, C#, Ruby, SQL, GraphQL

Full Stack & Backend: React, React Native, Next.js, Node.js, Express, PostgreSQL, WebSockets

Cloud & Infrastructure: AWS (Bedrock, S3, Nova), Docker, Terraform, GitHub Actions, CI/CD, Git

AI & Specialized: RAG (Retrieval-Augmented Gen), Vector Search, Android SDK, ARKit